SAMPLE

Relativity

Einstein's Special and General Theories

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Introduction

There are two theories of relativity, not one, and the fact that they share a name hides a story worth telling properly. The special theory, finished in 1905, is about light, motion, and the surprising behavior of clocks and rulers. The general theory, finished in 1915, is about gravity, and it reaches much further: into the lives of stars, the shape of the universe, and the strange objects we now call black holes. Between those two dates lies a struggle of ten years, conducted almost entirely inside one man's head, that turned a neat result about light into a new picture of what space and time actually are.

This book tells the story of both theories as one connected journey. Most introductions slice the subject in half, presenting special relativity as a chapter of physics and general relativity as a topic for specialists, and the connection between them gets lost. But the connection is the best part. The general theory grew directly out of a question the special theory left dangling, a single question about a falling person that a physicist asked himself in 1907 and spent eight years answering. When the answer arrived, it had swallowed the original question whole and rebuilt the universe around it.

The journey here runs from the laboratory failures of the 1880s, when physicists could not detect the medium they were sure light needed, through the miraculous spring of 1905, across the quiet decade of mathematics that followed, and into the dramatic November of 1915 when the final equations appeared in a rush of lectures delivered on four consecutive Thursdays. After that come the tests: an anomaly in Mercury's orbit, a total eclipse photographed from two hemispheres, a universe that refused to sit still, and, much later, ripples in space itself caught crossing the Earth a century after they were predicted.

No equations appear in these pages, because none are needed to follow the argument. What the theories say can be said in plain sentences, and saying it that way has a benefit the textbooks rarely admit: stripped of its mathematics, relativity becomes a story about the courage to question the most obvious things, the things so basic that nobody had thought to check them. What is time? What is distance? What does it mean to fall? A clerk in a Swiss patent office decided these were open questions, and the twentieth century spent the next hundred years confirming that he was right to ask.

A note on how the chapters are arranged, since the book is deliberately shaped like a journey rather than a reference. The first six chapters build the special theory and the motivation for something larger; the middle chapters carry the long construction of the general theory and its first victories; the closing chapters follow the theory's consequences out to collapsed stars, waves in space, and the readers' own pockets. Along the way, the historical order of discovery is kept wherever it explains the physics, and departed from wherever it obscures it, and each chapter opens and closes on its own argument so that a reader can stop anywhere without losing the thread. Nothing here requires a scientific background, only patience with sentences that deserve a second read.

The chapter's drama deserves a note on its scale, because the stubbornness was measured in the third decimal place: the ether experiments, ingenious and patient, designed to catch the drift of the planet through the medium, the answer, always the same null result, the light, moving at its one speed regardless of the observer's motion, the failure, the most productive in physics. The reader who thinks science advances by successes can meet the counterexample, the null, repeated and refined for decades, the anomaly, the door the theory walked through, the account, begun with a wrong answer that refused to budge.

And the chapter should name the stubbornness's other half, the measurement that made the null respectable: the light's speed, measured with increasing care by generations of experimenters, the number, always the same to the limits of the instruments, the invariance, the one fact every theory was invited to explain, the answer, delayed for decades because it required letting go of time itself. The reader who has held one fact against a hundred intuitions can hold this one, the constancy, the theory's only axiomatic demand, the courage, not in the postulate but in the obedience to it, the account, begun with the measurement nobody could bend.

Chapter 1: Light That Refused to Cooperate

Every good scientific story starts with an embarrassment, and the embarrassment here was light by the nineteenth century's middle years, physicists had achieved a magnificent description of light as a wave. Waves, as everyone could observe in water and sound, need something to wave in. Sound uses air; ripples use water. Light, which crosses the emptiness between the stars, must use something too, and the physicists of the age gave that invisible something a name: the luminiferous ether, a substance imagined to fill all of space, stiff enough to carry the fastest

waves known and insubstantial enough to pass unnoticed through every hand and every wall.

The only problem was that the ether refused to appear in any experiment designed to find it. The most famous of those experiments belongs to Albert Michelson, an American physicist of legendary experimental skill, and Edward Morley, his chemist collaborator in Cleveland. Their reasoning was straightforward. If the Earth rides through the ether as it orbits the Sun, then light should travel at slightly different effective speeds in different directions, just as a swimmer crosses a river faster going with the current than across it. In 1887 they built an instrument sensitive enough to detect that difference by letting two beams of light race each other along perpendicular paths and comparing their times on recombination. The race was rigged to reveal the Earth's motion through the ether. They ran it at different times of day, in different seasons, expecting the Earth's orbital motion to betray itself.

The result was silence. The two beams returned as if the laboratory were standing perfectly still in the ether, not hurtling through it at thirty kilometers a second. However the apparatus was oriented, light behaved as though the Earth were at rest, which was either a miracle or a message. Michelson, who had built his career on precision, considered the experiment a failure. Others were not so sure.

Among the people who took the null result seriously were two of the finest theoretical physicists of the era, the Dutchman Hendrik Lorentz and the Frenchman Henri Poincaré. Lorentz found that if moving objects contracted slightly in their direction of motion, as measured by an observer at rest, and if moving clocks ran slightly slow, then the mathematics of light waves could be made consistent with the Michelson-Morley result. It was a patch, and a strange one: rulers really shrinking because they moved, atoms pressing closer together as they traveled. Poincaré, who thought more deeply about the principles underneath, gave a lecture in 1904 arguing that a law of nature more general than any yet formulated seemed to be at work, a principle of relativity according to which no experiment performed inside a uniformly moving laboratory could ever reveal its speed. The pieces of a revolution were lying on the table, assembled by two masters who could not quite see what they had built, because each kept trying to save the ether. What nobody had yet suggested was that the ether simply did not exist, and that the strange shrinkings and slowings were not tricks played by matter moving through a medium but honest facts about space and time themselves. That suggestion required someone willing to throw away the furniture entirely, and in 1905 it arrived, from the least likely address in physics.

It is worth pausing on what a failed experiment can teach, because the lesson shaped everything that followed. Physicists of the era treated an experiment's silence as an embarrassment to be engineered around, and the engineering was ingenious: the ether was allowed to be dragged along by the Earth, wholly or partly, or to squeeze matter that moved through it, each patch buying the null result at the price of some new awkwardness. The alternative reading, which the nineteenth century could not quite bring itself to take, is that absence is data. An instrument built to detect a substance, refined by the best hands of the age, repeatedly finding nothing, is telling you something with the same authority as a positive detection, provided you are willing to consider that the substance itself is the flaw in the reasoning. The Michelson-Morley result sat in the literature for eighteen years as a curiosity before anyone drew the honest conclusion, and the delay is a study in how much easier it is to refine an instrument than to revise an assumption.

Before the null experiments of the 1880s, the ether had actually seemed to be working. Astronomers had long observed that the apparent positions of stars shift slightly through the year as the Earth moves, an effect called aberration, discovered accidentally in the 1720s by James Bradley while hunting for stellar parallax, and the shift's size and direction matched light behaving as a wave traveling through a medium past which the Earth moves. Optical experiments in flowing water by Hippolyte Fizeau in the 1850s added a subtlety, showing that light in moving water is neither fully dragged by the water nor fully independent of it, a partial cooperation that Lorentz's mathematics of a stationary ether could account for. The ether's credentials, in other words, were solid, which made its refusal to appear in the one experiment designed to trap its flow all the more maddening, and made the eventual verdict, that the medium was a habit of thought rather than a substance, all the more radical.

The patent office deserves its own paragraph as an institution, because the clerking was better training than it looked: the applications, streaming through the young examiner's desk, the electromagnetic patents, the problem of synchronizing distant clocks for the railways and the telegraphs, the day job, an education in exactly the puzzles of time and signal that the 1905 paper would solve, the office, the theory's unacknowledged classroom. The reader who pities the clerk can revise the pity, the post, secure and undemanding, the evenings, free for the real work, the synchronizing of clocks, Bern's own problem, the spring, seeded by the day job, the theory, born with a timekeeper's hands.